



الجمعية العلمية الطلابية
كلية الهندسة التكنولوجية | جامعة البلقاء التطبيقية

2018

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القرشي

مختبر دوائر كهربائية

لاب سيركت

الإمتحان النظري



طلاب البور والاتصالات



spark team

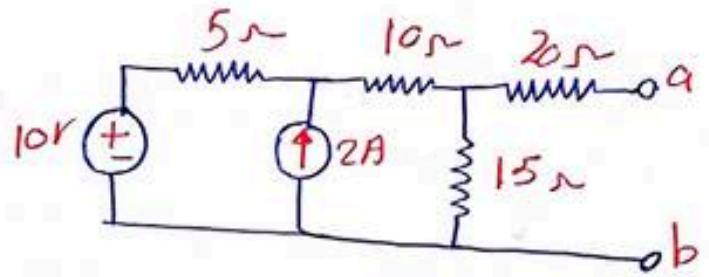


spark-fet.com



Q1: Find V_{th} , R_{th}

R_{th} : $2A \rightarrow$ open circuit
 $10V \rightarrow$ short circuit

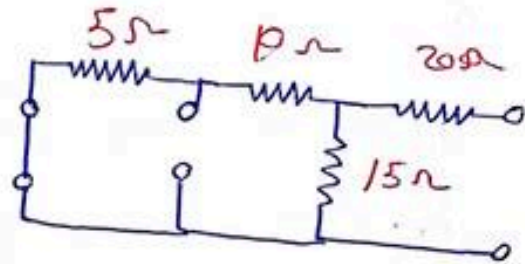


$$5 + 10 = 15\Omega$$

$$15 // 15 = 7.5\Omega$$

$$20 + 7.5 = 27.5\Omega$$

$$R_{th} = 27.5\Omega$$



V_{th} :

في هذه الحالة (10Ω) و (20Ω)
 في حالتي دوائر (0.5)

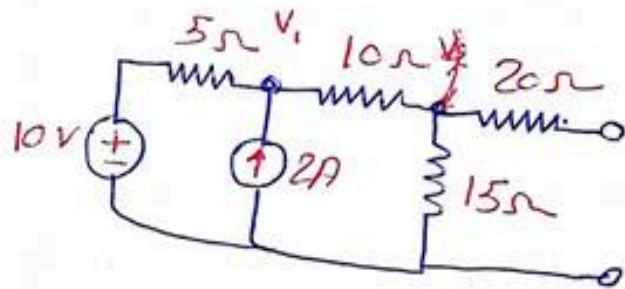
$$V_{th} = V_{15\Omega}$$

$$\frac{V_1 - 10}{5} + \frac{V_1}{10 + 15} = 2$$

$$V_1 \left[\frac{1}{5} + \frac{1}{25} \right] = 2 + 2$$

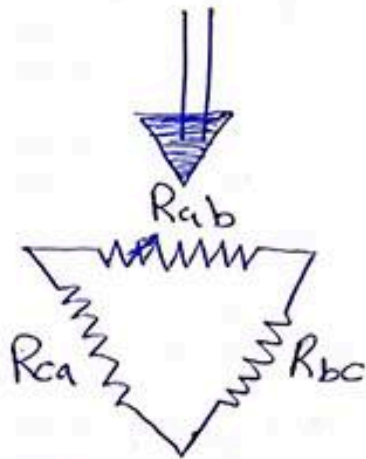
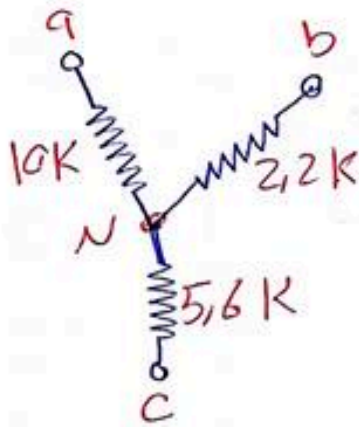
$$V_1 = 16.66V$$

$$V_{th} = \frac{V_1 * 15}{10 + 15} = 10V$$



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Q2: Transfer the connection $Y \rightarrow \Delta$:



$$R_{ab} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_c} = 16,13 K\Omega$$

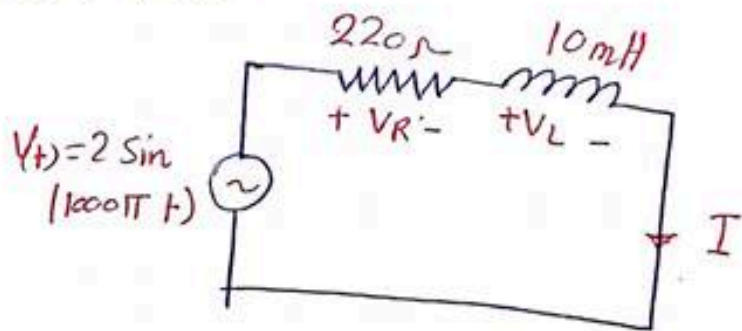
$$R_{bc} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_a} = 9,03 K\Omega$$

$$R_{ca} = \frac{R_a R_b + R_a R_c + R_b R_c}{R_b} = 41,05 K\Omega$$

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Q3: for the R-L circuit - find:

- ① f, T
- ② X_L, Z, I_{rms}
- ③ V_R, V_L
- ④ $\theta, P.F$



Sol: ① $2\pi f t = 1000\pi t \Rightarrow f = 500 \text{ Hz}$
 $f = \frac{1}{T}, T = \frac{1}{f} = \frac{1}{500} = 2 \text{ ms}$

② $X_L = j\omega L = j2\pi * 500 * 10 * 10^{-3} = j31,4 \Omega$
 $Z = R + jX_L = \sqrt{R^2 + X_L^2} = 222,23 \Omega$

$I_{rms} = \frac{V_{rms}}{Z} = \frac{2}{\sqrt{2} * 222,23} = 6,364 \text{ A}$

③

$$V_R = I_{rms} * R = 1,4 \text{ Volt}$$

$$V_L = I_{rms} * X_L = ~~0,1~~ 0,2 \text{ Volt}$$

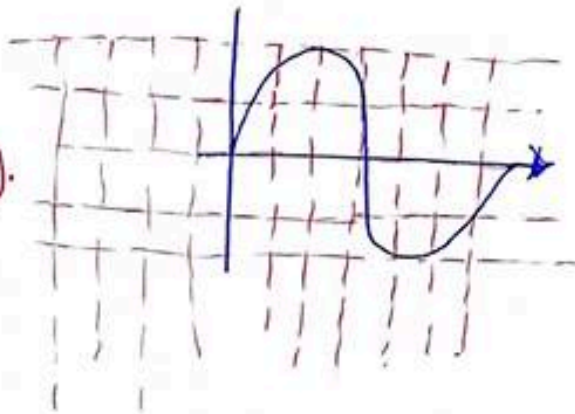
④

$$\theta = \tan^{-1} \left(\frac{X_L}{R} \right) = 8,123^\circ$$

$$P.f = 0,143$$

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Q4: Vertical sensitivity is
equal (50mV/div) and
Horizontal sensitivity (0,2μs/div).



Find:

- ① ~~Re~~ V_p , f .
- ② Read Voltmeter.
- ③ Find the division when $V_{pp} = 200 \text{ mV}$.
- ④ Find the division when $T = 10 \mu\text{s}$.
- ⑤ when $f = \infty$ in RC circuit find, V_c , P.f.

Sol:

① $V_p = 3 * 50 \text{ mV} = 150 \text{ mV}$

$$T = 6 * 0,2 \mu\text{s} = 1,2 \mu\text{s}$$

$$f = \frac{1}{T} = \frac{1}{1,2 \mu\text{s}} = 833,33 \text{ KHz}$$

② $V_{rms} = \frac{V_p}{\sqrt{2}} = 106,07 \text{ Volt}$

③ $200 \text{ mV} = \frac{200}{150} * \frac{150}{50} \Rightarrow \frac{200}{50} = 4$

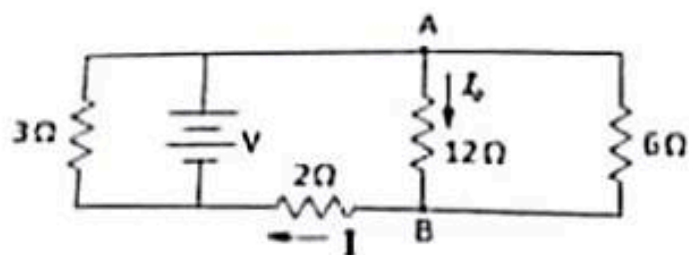
④ $10 \mu\text{s} = 0,2 \mu\text{s} * m \Rightarrow m = \frac{10}{0,2} = 50$

⑤ $f = \infty \quad X_L = 0 \quad V_c = \text{zero}$
 $P.f = \frac{0}{R} = 0$

③

For each question, choose the correct answer

Consider the circuit shown below, where $I_0 = 2\text{ A}$.



-The equivalent resistance seen by the source (R_{eq}) is:

- a. $23\ \Omega$ b. $2\ \Omega$ c. $16\ \Omega$ d. $5\ \Omega$

-The value of the voltage V_{AB} is:

- a. 24 V b. 6 V c. 4 V d. 48 V

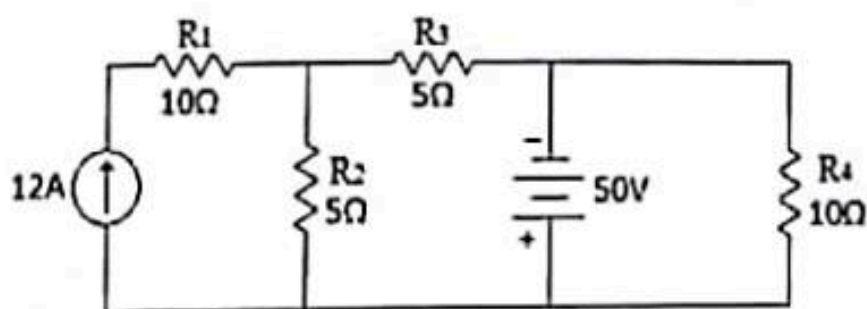
-The value of the current I is:

- a. 1 A b. 10 A c. 3 A d. 6 A

-The value of the source voltage V is:

- a. 16 V b. 100 V c. 24 V d. 36 V

Consider the circuit shown below.



-The current through the resistor R_4 is:

- a. 1 A b. 11 A c. 5 A d. 12 A

-The current through the resistor R_3 is:

- a. 1 A b. 11 A c. 5 A d. 12 A

-The power absorbed by resistor R_2 is

- a. 5 W b. 125 W c. 605 W d. 720 W

-In Superposition theorem, while considering a source, all other voltage sources are:

- a. open circuited b. short circuited c. change its position d. removed from the circuit

-In Superposition theorem, while considering a source, all other current sources are:

- a. open circuited b. short circuited c. change its position d. removed from the circuit

Q1:

المسألة الأولى:

①

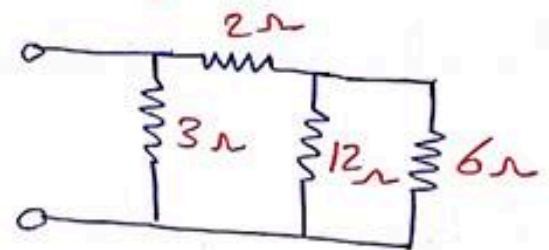
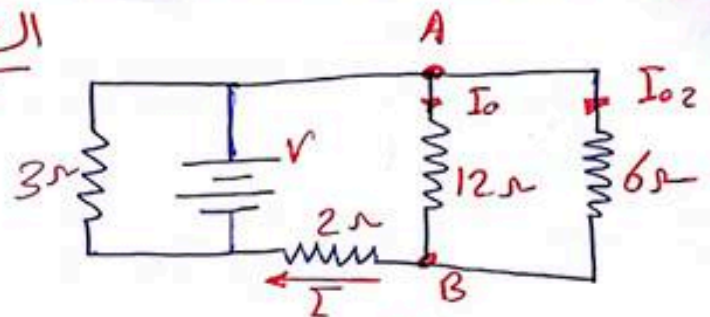
$$R_{eq} = ??$$

$$6\Omega // 12\Omega = 4\Omega$$

$$2\Omega + 4\Omega = 6\Omega$$

$$6\Omega // 3\Omega = R_{eq}$$

$$\boxed{R_{eq} = 2\Omega} \quad (b)$$



② $I_{01} = 2A$

$$V_{12\Omega} = I_0 \times 12\Omega = 2 \times 12 = 24 \text{ Volt} \quad (a)$$

③ $I_{02} = \frac{24}{6} = 4A$

$$I = I_{01} + I_{02} = 6A \quad (d)$$

④ $V = 6\Omega \times 6A = 36 \text{ Volt} \quad (d)$

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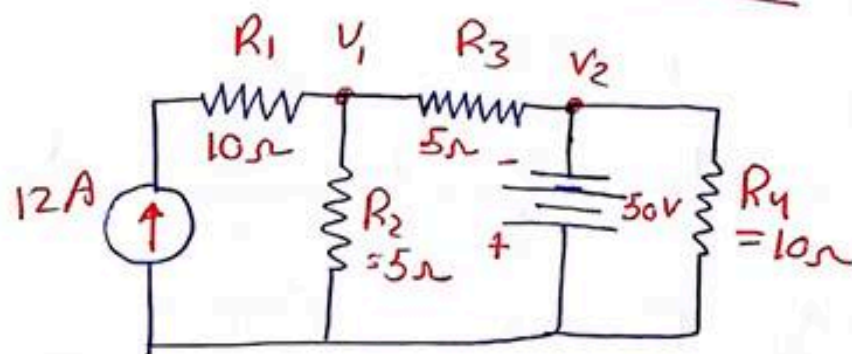
Q2:

① $I_{R4} = \frac{50}{10} = 5A \quad (c)$

② $\frac{V_1}{5} + \frac{V_1 - V_2}{5} = 12 \quad \dots (a)$

$$V_2 = -50 \text{ Volt}$$

$$V_1 = \frac{12 \times 5 - 10}{\left(\frac{1}{5} + \frac{1}{5}\right)} = 5 \text{ Volt}$$



$$I_{R3} = \frac{V_1 - V_2}{5} = \frac{5 + 50}{5} = 11A$$

(b)

(1)

-Which of the following series combinations dissipates the most power when connected across a source:

- a. One 100Ω resistor b. Two 100Ω resistors c. Three 100Ω resistors d. Four 100Ω resistors

-If one of the resistors in a parallel circuit is removed, the total resistance:

- a. Decreases by the value of the removed resistor
b. Remains the same
c. Increases
d. Non of the above

-The Thevenin equivalent voltage for a given circuit is found by:

- a. Shorting the output terminals
b. Opening the output terminals
c. Shorting the voltage source
d. Opening the voltage source

-When measuring current drawn from a DC power supply, it is true to say that the meter will act in circuit as:

- a. A perfect conductor b. An open resistance c. An insulator d. An extra current drain

-What does a multimeter measure?

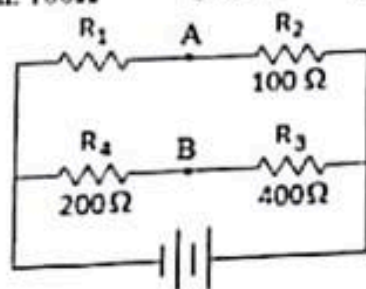
- a. Resistance, capacitance and inductance
b. Voltage, current and resistance
c. Resistance and reactance
d. SWR and power

-In measuring volts and amperes, the connections should be:

- a. The voltmeter in series and ammeter in parallel
b. The voltmeter in parallel and ammeter in series
c. Both voltmeter and ammeter in parallel
d. Both voltmeter and ammeter in series

-For the circuit shown below, $V_{AB}=0V$, the value of R_1 :

- a. 100Ω b. 50Ω c. 200Ω d. 800Ω



-When a sine wave has a frequency of 60Hz . In 10sec it goes through:

- a. 6 cycles b. 10 cycles c. $1/16$ cycle d. 600 cycles

$$(3) P_{R_2} = \frac{V^2}{R_2} = \frac{(5)^2}{5} = 5 \text{ watt (A)}$$

(4) Short circuit (b)

(5) open circuit (a)

① If one of the resistor in a Parallel circuit is removed the (R_T):
 - increases. (c)

② The Thevenin equivalent voltage for a given circuit is found by:

(b) opening the output terminal.

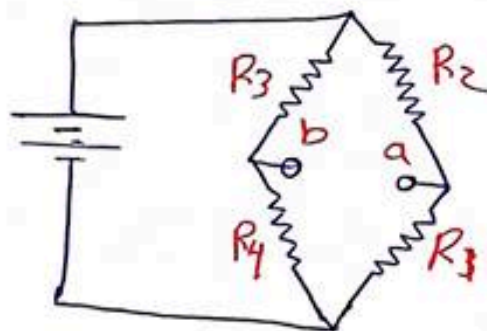
(3) (a) A perfect conductor.

(4) (b) Voltage, current and resistance.

(5) (b) The Voltmeter parallel and ammeter in series.

$$(6) \frac{R_3}{R_4} = \frac{R_2}{R_1}$$

$$R_1 = \frac{R_2 * R_4}{R_3} = 50 \Omega \text{ (b)}$$



$$(7) f = 60 \text{ Hz}, T = 10 \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{60} \text{ s} = T = 16.67 \text{ ms}$$

$$10 = 16.67 \text{ ms} * n$$

$$n = \frac{10}{16.67 \text{ ms}} = 600 \text{ cycle}$$

(2)

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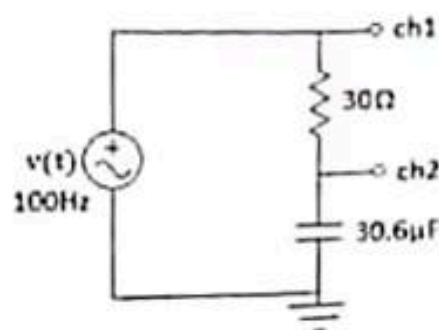
-A sinusoidal voltage is applied across an inductor, when the frequency of the voltage is increased, the current:

- a. Decreases b. Increases c. Remains constant d. Non of the above

-A capacitor and a resistor are connected in series to a sine wave generator. The frequency is set so that the capacitive reactance is equal to the resistance and thus, an equal amount of voltage appears across each component. If the frequency is decreased:

- a. $V_R > V_C$ b. $V_C > V_R$ c. $V_R = V_C$

Consider the circuit shown below:



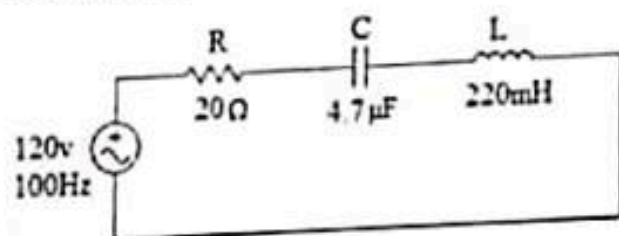
-The measured angle from both ch1 and ch2 is between:

- a. V_i and V_c b. V_i and V_R c. I_i and V_c d. I_i and I_c

-The phase shift for the circuit is:

- a. 60° b. 30° c. 45° d. 90°

Consider the circuit shown below:



-The amplitude of the impedance Z is:

- a. 201.63Ω b. 20Ω c. 496.96Ω d. 100Ω

-The power factor is:

- a. 0.099 lagging b. 0.099 leading c. 0.04 lagging d. 0.04 leading

-Total current is:

- a. 0.24A b. 6A c. 0.6A d. 12A

-Voltage across L is:

- a. 120v b. 828.9v c. 33.2v d. 82.9v

-The resonance frequency for the circuit is:

- a. 100Hz b. 200Hz c. 983.4Hz d. 156.5Hz

$$\textcircled{8} \uparrow X_L = j2\pi f L \uparrow \quad V_C \uparrow$$

$$I = \frac{V_C}{X_L} = \text{constat} \quad \textcircled{C}$$

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$$\textcircled{9} \quad \cancel{V_R = I * R} \quad \uparrow V_C = I * X_L = I * \frac{1}{2\pi f C} \downarrow$$

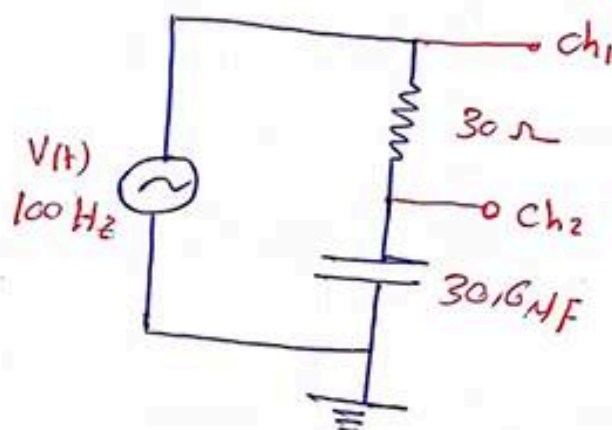
$$V_C > V_R \quad \textcircled{b}$$

Q3:

$$\textcircled{1} \quad V_f \text{ and } V_C \quad \textcircled{a}$$

$$\textcircled{2} \quad \theta = \tan^{-1} \left(\frac{X_C}{R} \right) = 60^\circ \quad \textcircled{a}$$

$$X_C = \frac{1}{2\pi f C} = 52,04 \, \Omega$$

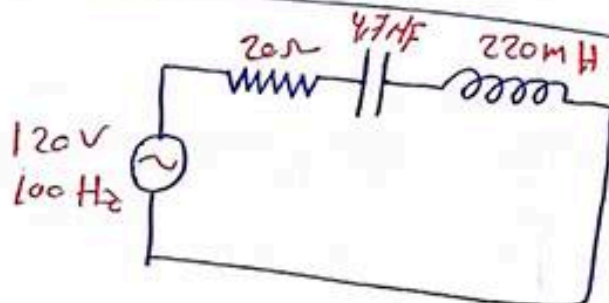


Q4:

$$\textcircled{1} \quad X_L = j2\pi f L = j138,16 \, \Omega$$

$$X_C = \frac{-j}{2\pi f C} = -j338,8 \, \Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = 201,63 \, \Omega \quad \textcircled{a}$$



$$\textcircled{2} \quad \cancel{\theta = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)}$$

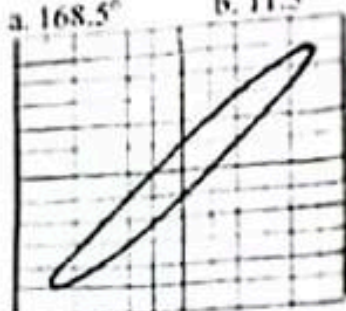
$$\cancel{P_f = \frac{V_R}{Z} = \frac{120}{201,63} = 0,595}$$

$$I = \frac{V}{Z} = \frac{120}{201,63} = 0,595 \, \text{A} \quad \text{lead}$$

$$P.f = \cos(84,1^\circ) = 0,999 \text{ leading} \quad \textcircled{b}$$

-The phase shift for the figure below:

- a. 168.5° b. 11.5° c. 78.5° d. 30°

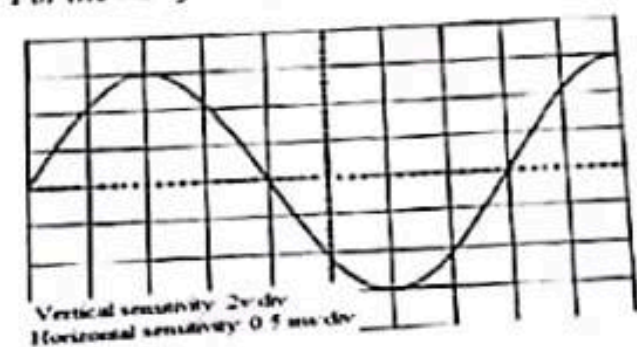


-The phase shift for the figure below:

- a. 38.5° b. 154.3° c. 77.1° d. 57.1°



For the waveform shown below answer questions



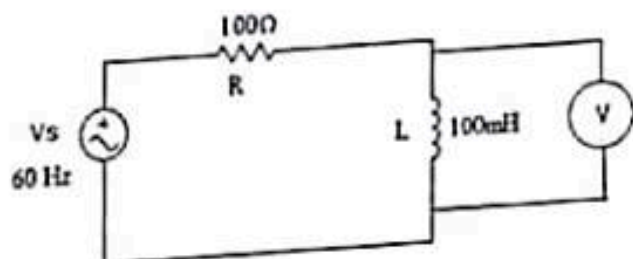
-The peak amplitude is (.....)

-The time period is (.....)

-The frequency is (.....)

-The RMS value is (.....)

For the circuit shown below the reading of voltmeter is 5V



- The voltage across resistance is:

- a. 5V b. 13.27V c. 14.18V d. 18.27V

- The source voltage is:

- a. 5V b. 13.27V c. 14.18V d. 18.27V

$$③ I_t = \frac{V}{|Z|} = \frac{120}{201,63} = 0,6 \text{ A } ③$$

$$④ V_L = I_t * X_L = 82,9 \text{ V } ④$$

$$⑤ f_c = \frac{1}{2\pi \sqrt{LC}} = 156,5 \text{ Hz } ④$$

السعة الرابطة

$$① \theta = \sin^{-1} \left(\frac{Y_n}{Y_m} \right) \\ = \sin^{-1} \left(\frac{2}{10} \right) = 11,5^\circ ⑥$$

$$② \theta = \frac{P_2}{P_1} * 36^\circ = \frac{3}{14} * 36^\circ = 77,1^\circ ③$$

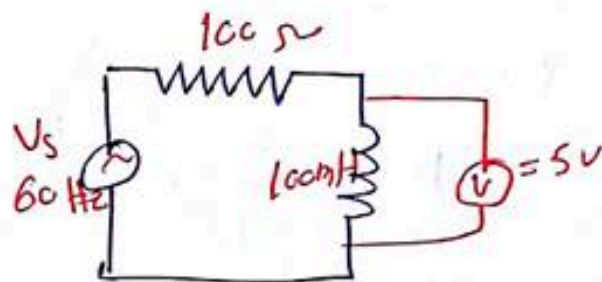
Q5:

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- $V_p = 2 * 4 = 8 \text{ Volt}$
- $T = 0,5 \text{ ms} * 8 = 4 \text{ ms}$
- $f = \frac{1}{T} = 250 \text{ Hz}$
- $V_{r.m.s} = \frac{V_p}{\sqrt{2}} = 5,66 \text{ Volt}$

Q6:

$$① X_L = j2\pi f L = 37,68 \text{ } \cancel{\Omega} \\ I_{r.m.s} = \frac{5}{\sqrt{37,68^2 + 100^2}} = 46,79 \text{ mA}$$



$$V_R = R * I_{r.m.s} = 4,68 \text{ Volt} \approx 5 \text{ V } ④$$

$$② V_s = \sqrt{V_R^2 + V_L^2} = \sqrt{5^2 + 5^2} = 5 \text{ Volt } ④$$